Dot Net Full Stack Web Development

Course Introduction

Alternate frameworks & technologies available for Project Development

Why Dot Net?

What is Dot Net?

Technologies & Languages Used in Dot Net

SQL

SQL Fundamentals

- SQL Introduction
- RDBMS
- SSMS
- Connect To SQL Server

SQL COMMANDS

- Database
- Table
- Columns
- Table constraints
- Cascading referential integrity

Query Operations

- Select statement
- Where clause
- Order by clause
- Group by clause
- Having clause
- Aggregate functions
- Identity

Advanced Queries

- Subquery
- Join
- Replace null value
- Union
- View
- Temporary table
- Table variable
- select into command
- Relationships

Programming Elements

- Stored Procedures
- Conditional statements
- looping statement
- Function
- String functions
- Conversion functions
- Mathematical functions
- Deterministic function
- Non deterministic function

Advanced Features

- Date in SQL
- Functions to retrieve current datetime
- Date related functions
- Date from parts functions
- User defined functions
- Inline table valued function
- Multi statement table valued function

Database Objects

- Trigger
- Magical tables
- DDL Triggers
- Derived table
- CTE

Performance & Management

- Index
- Cursor
- Error handling
- Raiserror() function
- Transaction

Advanced Concepts

- Isolation Level
- ACID property
- Ranking functions
- Normalization
- Pivot
- Unpivot

C#

C# Fundamentals

- C# introduction
- IDE
- Project
- .cs file
- Build option
- Clean Option
- Rebuild option

Data & Variables

- Datatypes
- variable
- String functions
- Datetime datatype
- Datatype conversion
- Operators



Control Flow

- Conditional Statements
- Switch statement
- Nullable type
- Type of datatypes
- Boxing operation
- Unboxing operation



Loops & Arrays

- Looping statements
- Initialization
- Condition
- Updation
- Break statement
- Continue statement
- Array
- Foreach loop

Methods

- Method
- Optional parameter
- Different types of Method parameters
- Calling method
- Called method

OOP Basics

- Class
- Static keyword
- Static class
- Constructor
- Inheritance
- This keyword
- Base keyword
- Constructor behaviour in inheritance

OOP Advanced

- Method hiding
- Polymorphism
- Method overriding
- Method overloading
- Operator Overloading
- Namespace
- Reusable project/ assembly

Access & Abstraction

- Access tree
- Access Specifiers
- Abstract keyword
- Abstract class
- Multiple class inheritance
- Interface
- Why interface?
- Difference between abstract class and interface
- Abstraction
- Encapsulation
- How we implement abstraction & encapsulation
- Association

Properties & Exception Handling

- Property
- Access modifiers
- Type of properties
- indexer
- Exception
- Exception Handling
- Exception classes
- What do we mean by handling exception?
- What will happen if exception occurs in catch block?
- To throw exception intentionally
- Finally block

Advanced Types

- Object type
- Var keyword
- Dynamic keyword
- Sealed keyword
- Structure or struct
- Record
- Enum

Functional Programming

- Generics
- Delegate
- Anonymous method
- Generic delegate
- Lambda Expression

Miscellaneous Features

- Const & readonly keyword
- Is as keywords
- Partial class
- Partial method
- Collections
- C# provides collection classes
- Tuple
- Yield keyword
- Lazy Loading
- Extension Method
- Reflection
- Attribute
- Association

HTML

HTML Introduction

Basic Elements

- Line breaks
- Html page structure
- Heading tags
- p tag
- pre tag

Media Elements

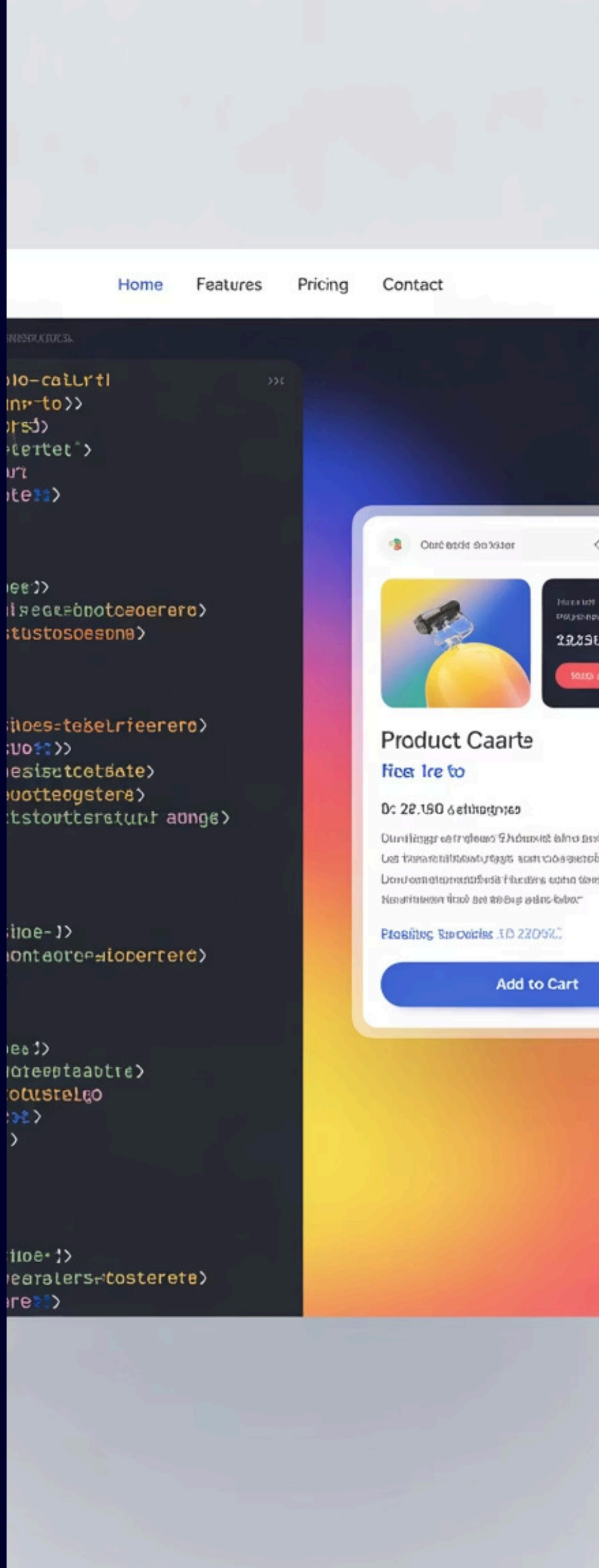
- img tag
- a tag
- Video tag
- Audio tag

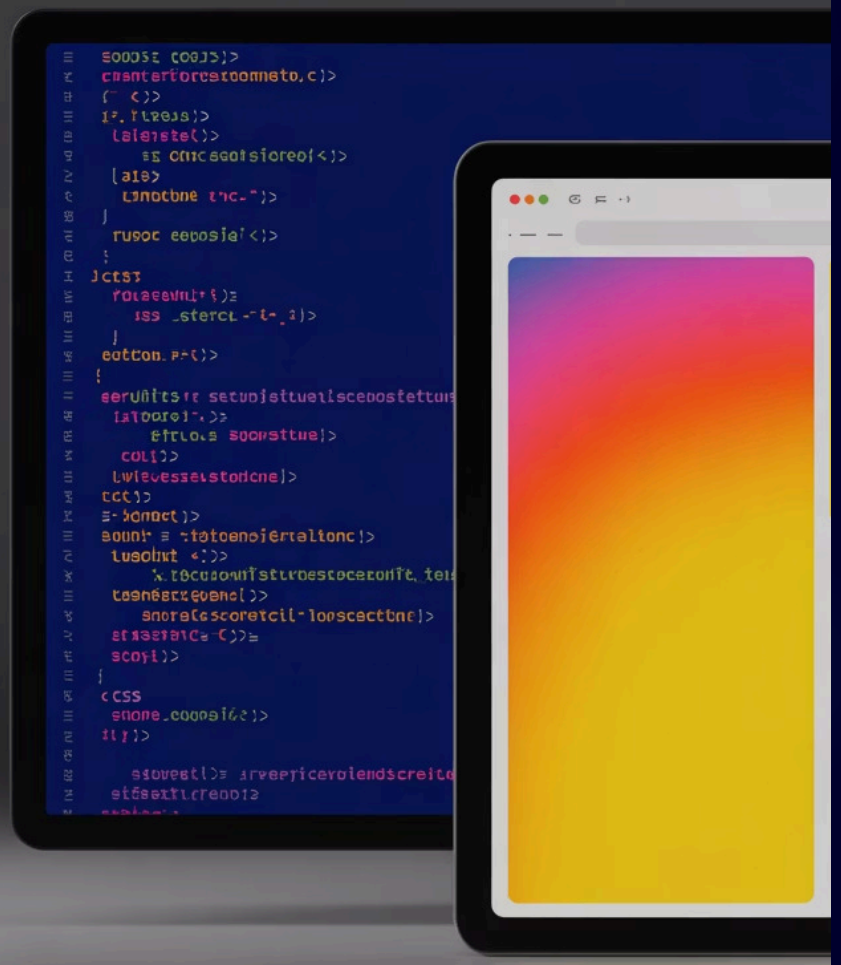
Form Elements

- Label tag
- Form tag
- Input fields/ controls using input tag
- Button
- Textarea tag
- Select tag

List & Container Elements

- Ul tag
- Ol tag
- Container tags
- Table tag
- Iframe tag





CSS

CSS Introduction

To apply CSS on html page

Selectors

CSS Selector

Visual Properties

Color

Box Model

Border

Margin

Padding

Visibility

Opacity

Visibility & display

JAVASCRIPT

JavaScript Introduction

Fundamentals

- DOM
- Datatypes
- Applying Javascript on html page
- To show output on browser
- String datatype
- Operators

Control Flow

- Conditional statements
- Looping statements
- Break statement
- Continue statement

Arrays & Functions

- Array
- mutator functions
- non mutator functions
- Array functions
- functions in javascript
- undefined
- NaN
- arguments object

Function Types

- ways to declare functions in Javascript
- Function hoisting
- Type of variables
- Variable hoisting
- Closure function

Error Handling & Utilities

- Debugging Javascript Code
- Error handling in Javascript
- Date in Javascript
- Timing Events

DOM Manipulation

- Get set content from HTML page
- get set attribute
- Events
- Disabling browser default actions
- Ways to register or assign events
- event object
- identify which mouse button is clicked
- event bubbling
- event capturing

Advanced Concepts

- Place for script tag
- open() function
- Javascript Minification
- Browser Storage
- Regular Expression

JQUERY

Jquery Introduction

Jquery syntax

Refer jQuery

Ready function



Core Concepts

- Selectors
- jQuery effects
- Get set content
- Get set attribute



DOM Manipulation

- Add elements
- Remove elements
- get set css classes
- Get set style properties
- Dimensions



Traversal & Filtering

- Traversing
- Filtering functions
- $ in jQuery

Additional Libraries

- Datatables library
- jQuery UI
- moment js
- jQuery AJAX

BOOTSTRAP

Bootstrap Introduction

To include bootstrap in our project

Layout & Typography

- Grid system
- Units
- Typography
- Font colors

Content Elements

- tables
- Whenever application requirements comes from client
- Images
- Jumbotron

Components - Part 1

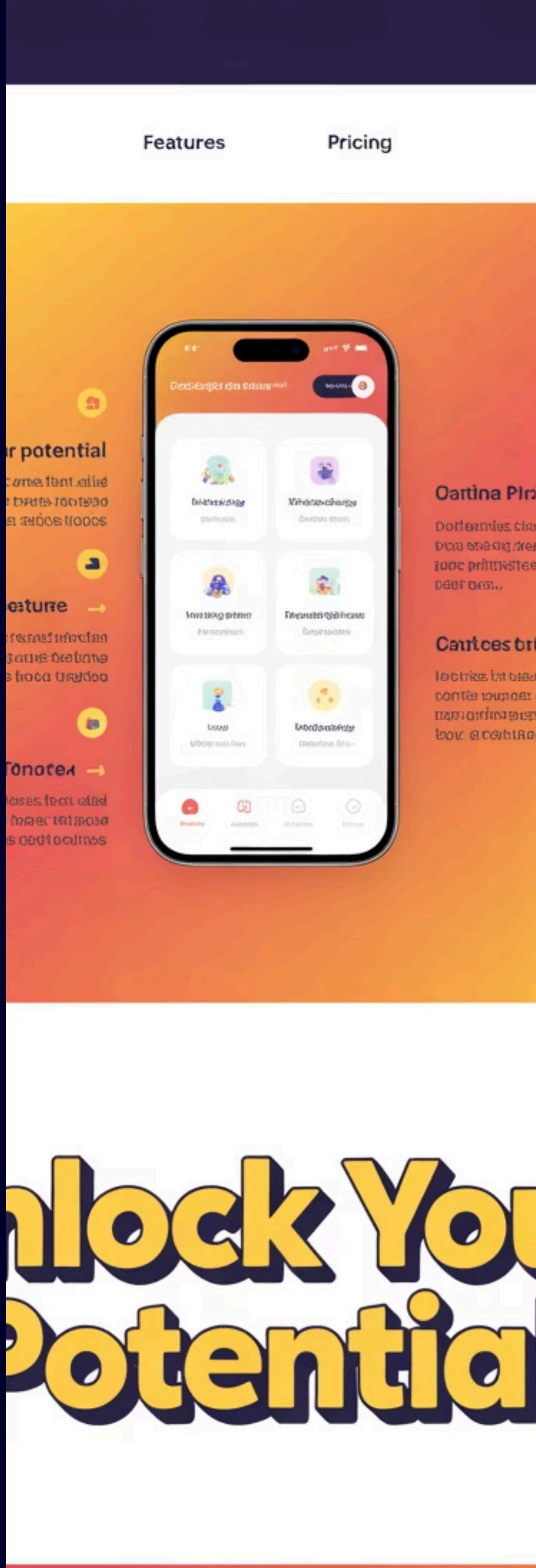
- Alerts
- Buttons
- Button Groups
- Badge
- Progress bar
- Spinner
- Pagination
- List group
- Cards

Components - Part 2

- Dropdowns
- Collapse
- Navs
- Navbar
- Carousel
- Modal
- Tooltip
- Popover
- Toast
- Offcanvas

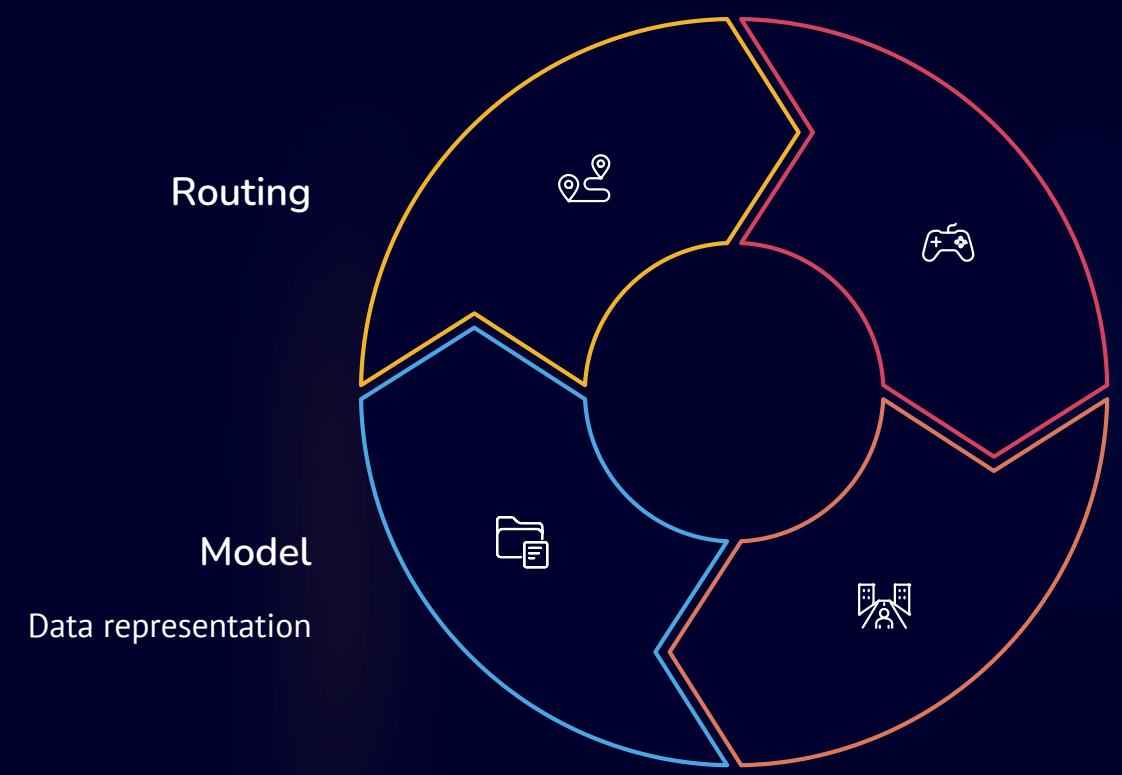
Utilities & Forms

- Utilities
- Flex
- Forms
- Input group



MVC

MVC Introduction



View Components

- HTML helper class
- Strongly typed view
- Views

Data Flow

- Passing data from controller to view
- Session State Variable
- Application State Variable
- Transfer Data From View To Controller
- To pass value using url
- Action Method Parameters

Controller Features

- Scaffolding in MVC
- Model Binder
- CRUD operations using Session Object
- ActionName attribute
- NonAtion attribute

Database Connectivity

- Database Connectivity or Communication
- ADO.NET
- Data Providers
- Connection string

ADO.NET Classes

- SqlConnection class
- SqlCommand class
- SqlDataReader class
- We can execute queries in two approaches
- Performing CRUD operations using stored procedures
- Calling stored procedure having output parameter
- Storing connection string
- SqlDataAdaptar
- Dataset

Entity Framework

- Entity Framework
- Entity Framework provides multiple approaches
- Database First Approach
- ADO.NET Entity Data Model Item Adds
- Entity surface
- Update Model From Database
- Code First Approach In Entity Framework
- Data Annotations

Validations

- Validations In MVC
- Validations example
- Validation attributes
- Both side validation
- Use of Metadata class
- Custom validation attribute
- Remote attribute
- Validation Summary

Advanced Features

- Uploading images or any file
- Type of paths
- Area
- Filters
- In built filters
- Output Cache
- Handle Error
- Custom Filter
- Applying Filter globally
- Custom action filters
- Bundling

SOLID PRINCIPLES

SOLID Principles

SRP = Single Responsibility Principle

A class should have only one reason to change, meaning it should have only one responsibility.

OCP = Open Closed Principle

Software entities should be open for extension but closed for modification.

LSP = Liskov Substitution Principle

Objects of a superclass should be replaceable with objects of its subclasses without affecting the correctness of the program.

ISP = Interface Segregation Principle

Many client-specific interfaces are better than one general-purpose interface.

DIP = Dependency Inversion Principle

High-level modules should not depend on low-level modules. Both should depend on abstractions.

Dependency Management

Dependency Injection (DI)

A technique where one object supplies the dependencies of another object.

IoC Containers

Frameworks that implement inversion of control for dependency injection.



DESIGN PATTERN

Design pattern

Pattern Categories

- Structural Design Pattern
- Creational Design Pattern
- Behavioral Design Pattern

Common Patterns

- Singleton Design Pattern
- Factory Design Pattern
- Repository Design Pattern

Project Architecture

- Multi Layer Project
- Multi Tier Project

Supporting Tools

- Auto Mapper
- Logging
- Application Logging
- Feature flag for Logging

WEB API

WEB API Introduction

Service Fundamentals

- Service
- Reusable code
- Protocol
- Message format

Service Types

- Services
- ASP.NET Web Service (asmx services)
- WCF Service
- WEB API

API Basics

- Web Api Project
- Routing
- Http Verbs
- Api method return type
- Http Status Code
- API Method Parameters

Implementation

- Implement CRUD Operations in WEB API
- Testing Web Api
- We use testing tools like
- Swagger To Test Api

Data Handling

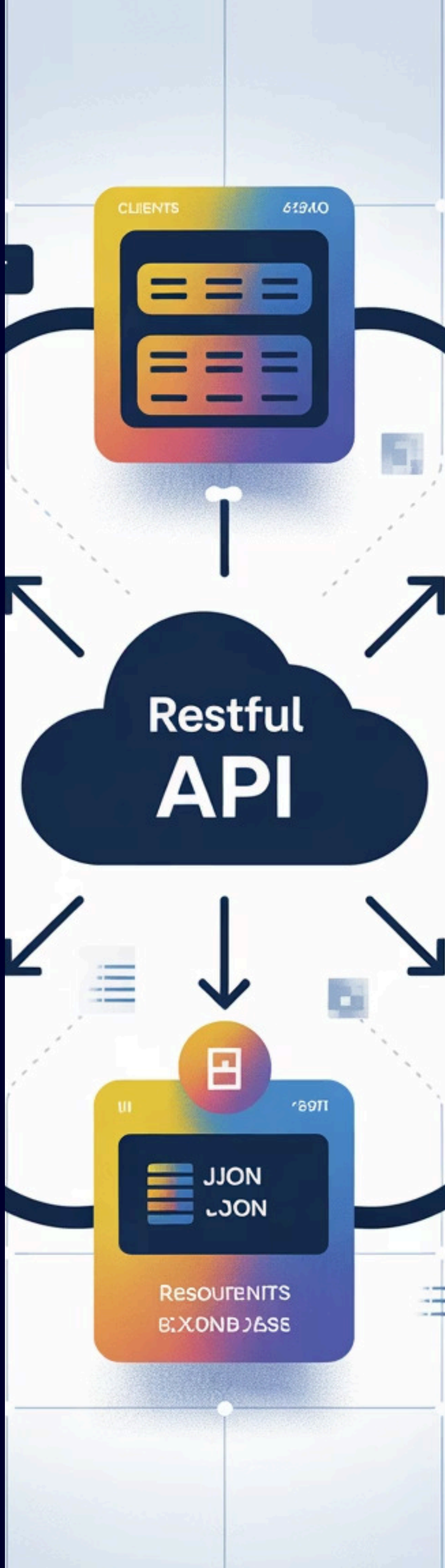
- Consuming API's
- Serialization
- De Serialization
- Serialization and Deserialization
- Content Negotiation/ Media Type Formatters
- CORS Policy
- Consuming API's using Javascript Ajax Call

Security

- Security in Api
- Authentication
- Token based authentication

Deployment

- Deployment
- IIS



DOT NET CORE

Framework Fundamentals

- Dot Net Framework
- Dot Net Framework Components
- Two Steps Compilation
- ILDASM
- Assembly
- GAC = Global Assembly Cache

Core Concepts

- Dot Net Core
- Garbage Collector
- Generation
- Resources
- Dispose() method
- Finalize() Method
- Assembly Parts

Environment Management

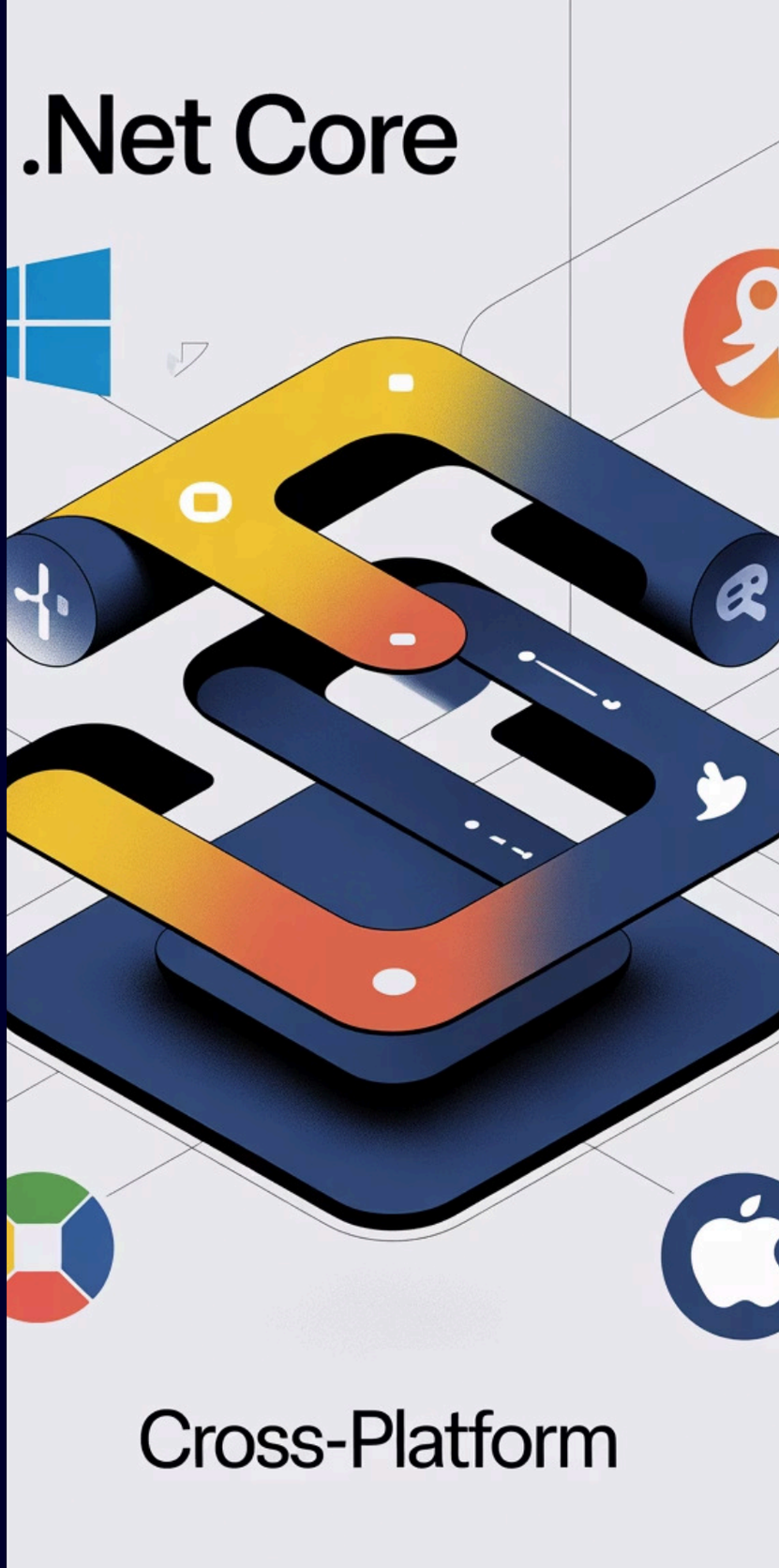
- Environment / all stages servers
- Type of environments

Testing Frameworks

- Testing
- Unit Testing
- MsTest
- Unit Testing Methodologies/ Techniques
- nUnit Testing Framework
- xUnit Testing Framework

Mocking

- Mocking Frameworks
- Available mocking frameworks



MVC CORE

Framework Overview

- MVC
- Dot Net Framework
- Dot Net Framework Components
- Two Steps Compilation
- ILDASM
- Assembly
- GAC = Global Assembly Cache
- Dot Net Core

Core Components

- Controller
- View
- View Component
- Routing
- Custom tag helpers
- Model Binder
- Transfer data from View To Controller

State & Caching

- State Management
- State Management Techniques
- Tempdata Providers
- Caching
- Response Caching

Data Access

- Database communication
- EF Core Code First Approach
- Multi layered project
- Automapper in MVC 8

Validation & Security

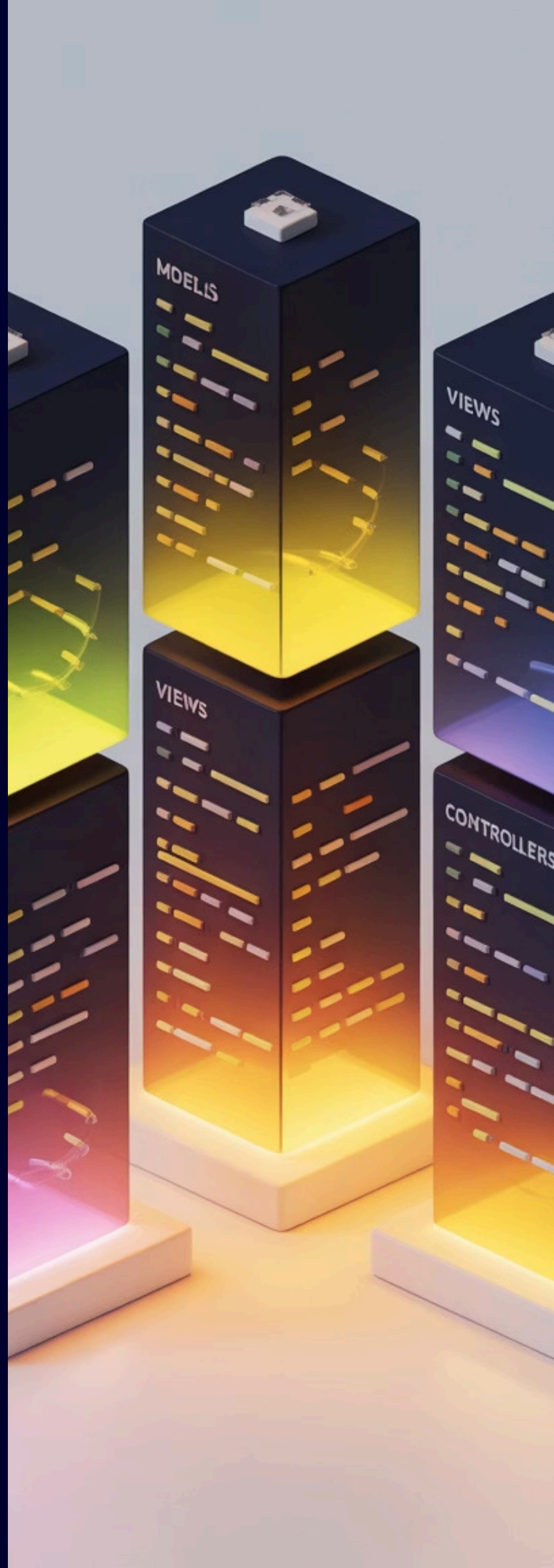
- Validations
- Authentication
- Authorization
- Encrypting Password

Application Architecture

- Registering dependency
- MVC Core 8 Request Lifecycle
- Filters
- Middleware

Testing & Logging

- Unit Testing
- Logging Framework



WEB API CORE 8

Why we use Web API

Why Web API Core?

API Fundamentals

- Http Verbs
- Http status code
- Model Binding
- [FromUri] attribute
- [FromBody] attribute
- CRUD Operations
- REST

Advanced Features

- Environment Variables
- Asynchronous API Methods
- Parameter binding
- Consuming Api's using HttpClient class
- Serialization and De serialization
- Caching
- Api Versioning

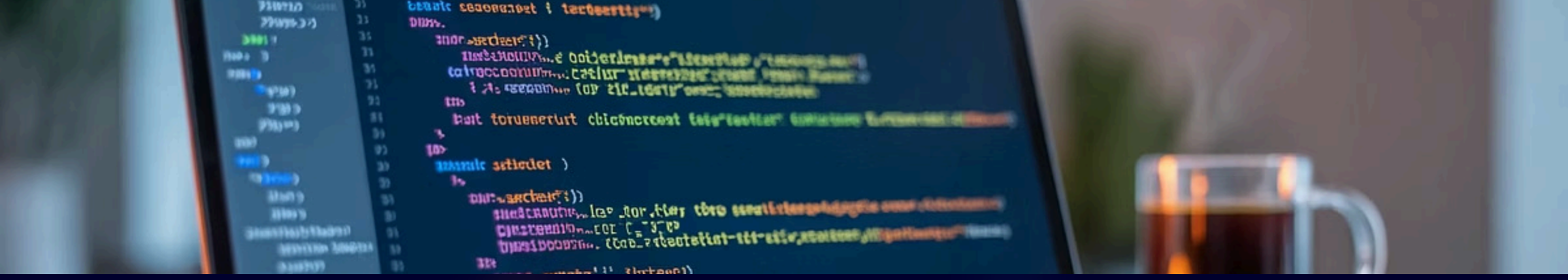
Security & Content Handling

Security

- Security
- Authentication and authorization
- Token based authentication
- OAuth

Content

- Content Negotiation



TypeScript

Pre requisites & Software's needed

- Node JS
- NPM
- TypeScript
- Visual Studio Code

Type System

- Datatypes
- Type Guard
- Type Inference & Assertion
- Array
- Tuple
- Enum
- Interface

Classes & Objects

- Class
- Class Object
- Class Members
- Constructor
- Method in detail
- Method Optional Parameters
- Method Rest Parameters

Control Flow

- Conditional Statements
- Looping Statements
- Continue, break and return statement

OOP Features

- Access Specifiers
- Constructor Parameter Property
- Static
- Abstract
- Inheritance
- Interface Implementation
- Method Overriding
- Method Overloading
- Generics

Angular

Getting Started

- Installing Angular
- Creating Angular Project
- Angular Project Structure
- Running Angular Project
- Component
- Template
- Data Binding

Directives & Pipes

- Directive
- Custom Directive
- Pipe
- Custom Pipe

Component Communication

- ViewChild, ViewChildren
- Content Projection
- ContentChild
- ContentChildren

Navigation & Services

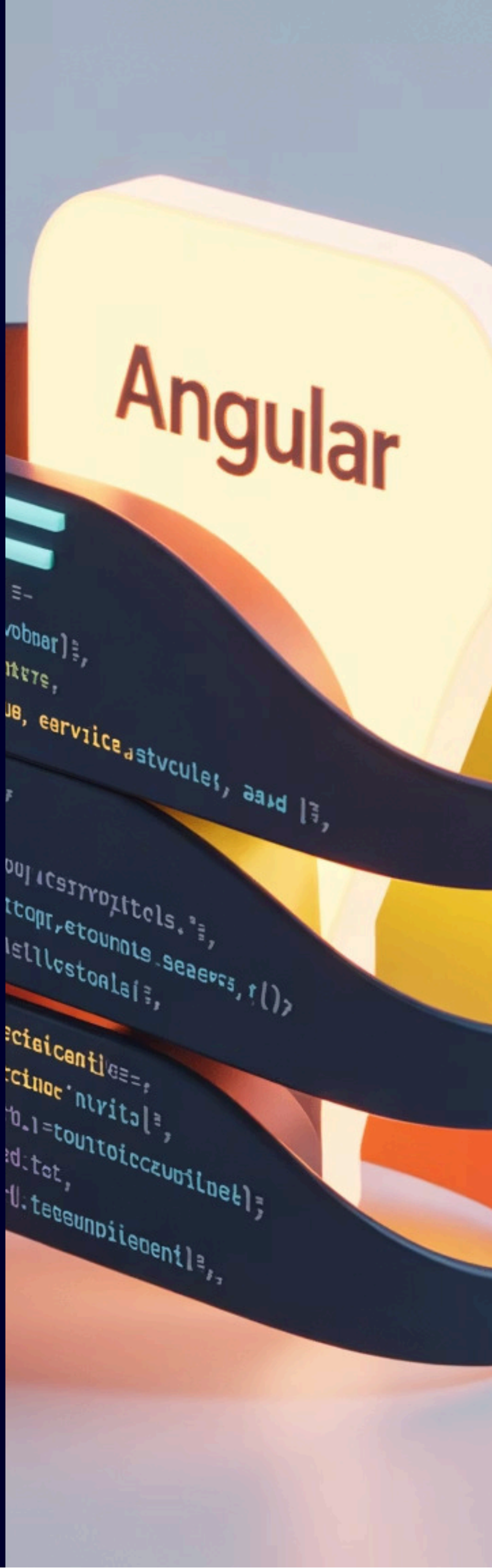
- Installing Bootstrap in Angular Project
- Routing
- Child Routing
- Route Guards
- Service
- Http Call from Angular

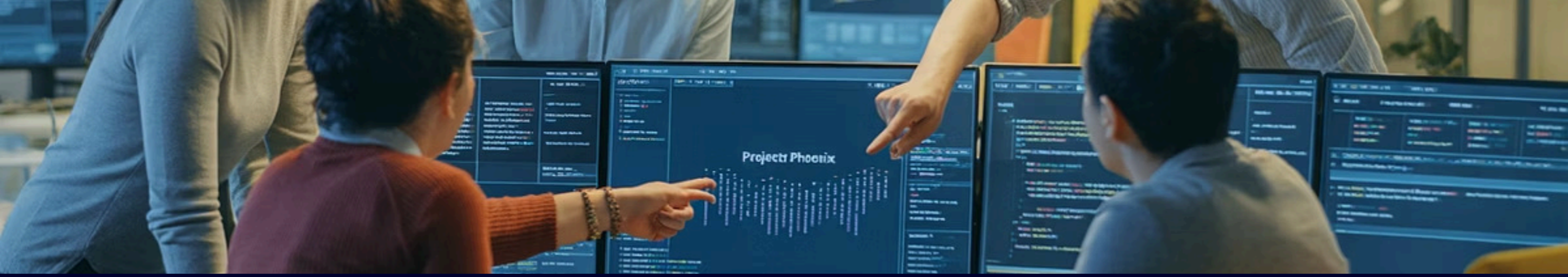
Architecture & Forms

- Module
- Lifecycle Hooks
- Forms in Angular
- Angular Material

Advanced Concepts

- Promise & Observable
- Http Interceptor
- Unit Testing in Angular
- Demo Angular Project for practice





Real Time Project, Processes, Tools & Work

Project Architecture

- Multi Layered Project
- Multi-Tiered Project
- Deployment using IIS

Development Tools

- Automapper
- Unit Testing using xUnit & MOQ Frameworks
- CICD using Jenkins
- SonarQube
- Nexus

Process & Practices

- Agile SDLC Process
- GIT & GITHUB
- Log4Net & Serilog Logging Framework



Resume Building

ATS & AI based resume creation

Learn how to create resumes that pass through Applicant Tracking Systems (ATS) and leverage AI tools to optimize your resume for job applications in the tech industry.

ATS Optimization

Techniques to ensure your resume passes through automated screening systems by using appropriate keywords and formatting.

AI Tools

Leveraging AI-based tools to create, refine and tailor your resume for specific job applications in the .NET ecosystem.

Mock Interviews

One or two mock interviews

Feedback shared for improvements

Interview Practice

Participate in realistic mock interviews that simulate actual job interview scenarios for .NET developer positions.

Personalized Feedback

Receive detailed feedback on your interview performance, including technical knowledge, communication skills, and areas for improvement.





Interview Assistance & Support

Share interview experience

Questions

Get support during interviews

Experience Sharing

Learn from the interview experiences of successful candidates who have secured positions in the .NET ecosystem.

Common Questions

Review frequently asked technical and behavioral questions specific to .NET Full Stack developer roles.

Ongoing Support

Receive guidance and assistance during your actual interview process, including pre-interview preparation and post-interview analysis.